OVERVIEW

Covariance averages the products of the two deviations from the means. Only its **sign** is read here: it reports the direction of the linear relationship, not its strength.

$$\operatorname{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

split by **sign of** $\operatorname{Cov}(X, Y)$

One number computed from paired observations. Its sign is read in three ways; its size on its own is not the message.

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CASE 01

$\operatorname{Cov} > 0$

Deviations move the same way

the mean product of deviations is positive

higher-than-average X tends to come with higher-than-average Y, and lower with lower

products $(X - \mu_X)(Y - \mu_Y)$ are mostly > 0

E.G. height and weight across people

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CASE 02

$\operatorname{Cov} < 0$

Deviations move opposite ways

the mean product of deviations is negative

higher-than-average X tends to come with lower-than-average Y

products $(X - \mu_X)(Y - \mu_Y)$ are mostly < 0

E.G. driving speed and trip time

0

CASE 03

$\operatorname{Cov} = 0$

No consistent linear relationship

the mean product of deviations is zero

the deviations show no steady direction; this does **not** mean the variables are independent

the products cancel out

E.G. $Y = X^2$ with X symmetric about 0: Y is fixed by X, yet the covariance is 0

TAKEAWAY

To compare strength across different pairs, divide by $\sigma_X \sigma_Y$: the correlation ρ_{XY} strips the units away and always lies in $[-1, 1]$.